

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-ae4f864cfa798fc0d.jsonl`  
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- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:20:06.579Z)

Source Extractor

Research question: "Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMM/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,

edit/Levenshtein score, MAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6716 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6716 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 375 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Fetch and extract key claims from this source:

****URL:**** <https://www.emergentmind.com/topics/temporal-action-segmentation>

****Title:**** Temporal Action Segmentation: evaluation-metric overview (segmental F1@k, edit/Levenshtein, frame-acc, MAP@tIoU)

****Found via:**** Small-n evaluation rigor for video segmentation search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-16T03:20:08.412Z)

I'll fetch the page content first.

3. user (2026-06-16T03:20:19.625Z)

Temporal Action Segmentation: Evaluation Metrics and Methods

****Last Updated:**** 12 January 2026

Evaluation Metrics

****Frame-wise Accuracy (MoF):****

"percentage of correctly labeled frames."

****Segmental Edit Score:****

"normalized Levenshtein distance between predicted and ground-truth segment sequences."

****Segmental F1@IoU:****

"F1 score over predicted vs. true segments, considering only those with temporal IoU exceeding thresholds (typically 10%, 25%, 50%)."

****Metric Sensitivity:****

"These metrics quantify both local classification accuracy and the coherence of temporal grouping, explicitly penalizing over-segmentation and boundary errors."

Key Methods and Performance Claims

****MS-TCN++**** (fully supervised baseline on 50Salads):

- Accuracy: 83.7%
- F1@50: 70.1
- Edit Score: 74.3

****Timestamp Supervision**** performance on 50Salads:

- Accuracy: 75.6% (vs. 83.7% full supervised)
- F1@50: 60.1 (vs. 70.1 full supervised)

****Gap Characterization:****

"timestamp supervision halves the gap to full supervision" and reaches within approximately 4-6% of supervised performance with reduced annotation burden.

****Unsupervised Methods**** (Breakfast dataset):

Best unsupervised mean-over-frames rate: 53% on Breakfast dataset.

4. assistant (2026-06-16T03:20:24.879Z)

The fetch succeeded with substantive, relevant content. Let me record the findings.

The source is EmergentMind, an AI-generated topic-overview aggregator that synthesizes research literature. It is a secondary source (it summarizes primary research like MS-TCN++ and timestamp-supervision papers) rather than a primary research paper. The content is directly on-target for dimensions 1 (windowing/boundaries metrics, over-segmentation) and 3/4 (timestamp supervision, small-data, evaluation methodology).

5. user (2026-06-16T03:20:33.845Z)

Structured output provided successfully

6. assistant (2026-06-16T03:20:44.557Z)

Recorded the findings. Summary of what this source contributes:

The source is ****https://www.emergentmind.com/topics/temporal-action-segmentation**** ? an AI-generated, citation-backed topic overview (secondary source, last updated 2026-01-12). It is relevant to research dimensions 1 (windowing/boundary metrics) and 3-4 (small-data supervision, evaluation rigor).

Key load-bearing extractions:

- ****Over-segmentation-aware metric triad****: frame-wise accuracy (MoF) + segmental edit score (normalized Levenshtein over segment sequences) + segmental F1@IoU at {0.10, 0.25, 0.50}. The source explicitly states these "penaliz[e] over-segmentation and boundary errors" ? directly validating the report's claim that plain tIoU/frame-accuracy is over-segmentation-blind and the edit/segmental-F1 pair is the standard fix.
- ****Timestamp supervision**** (point-supervised TAL, dimension 3) gives 75.6% acc / 60.1 F1@50 vs 83.7 / 70.1 for fully-supervised MS-TCN++ on 50Salads ? "halves the gap to full supervision" at low label cost.

- **Caveat on numbers**: the reported figures (MS-TCN++ 83.7%/70.1/74.3 on 50Salads; 53% MoF unsupervised on Breakfast) are plausible and consistent with the MS-TCN++ literature, but EmergentMind is an aggregator, not the primary paper ? the exact figures should be confirmed against the original MS-TCN++ / timestamp-supervision papers before being cited as primary evidence.

No license-relevant or contradictory claims surfaced. Structured output delivered.